Ali Razmdideh

Senior Machine Learning Engineer

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Summary

Experienced Machine Learning Engineer with over 9 years of professional experience, focused on applying AI technologies to simplify everyday life and drive meaningful global impact.

Profiles

Linkedin | Github | Stackoverflow

WhatsApp | Telegram

Education

Amirkabir University of Technology — Master of Science • GPA: 3.67/4.0

Control Engineering - Electrical Engineering — Tehran, Iran • Sep 2019 – Oct 2022 (While working full-time)

MSc Thesis – Deep Reinforcement Learning for Sequence-to-Sequence Lip Reading

Designed and implemented an end-to-end visual speech recognition system that maps sequences of lip movements to textual output using deep learning. Addressed robustness challenges across varying facial angles by generating multi-view facial data via GAN-based face rotation and augmentation techniques. Developed a reinforcement learning framework (policy gradient–based) to optimize training efficiency, dynamically control data sampling, and improve model generalization. Leveraged CNN/RNN/Transformer-based architectures for spatiotemporal feature extraction and sequence modeling, achieving improved accuracy and reduced training time on benchmark datasets (e.g., LRS2/LRS3). Integrated preprocessing pipelines including face detection, landmark extraction, and lip-region tracking to enable scalable real-world deployment.

University of Tabriz — Bachelor of Science • GPA: 3.39/4.0

Electrical Engineering — Tabriz, Iran • Sep 2015 – July 2019

BSc Thesis – Anomalous Particle Detection in Petrochemical Imagery

Developed a computer vision–based system to automate the inspection of petrochemical solutions by detecting and analyzing bubble patterns, replacing hazardous manual visual assessments. Implemented classical image processing techniques including **Canny edge detection** and **Hough Circle Transform** using **OpenCV** to identify and quantify bubble size, distribution, and irregularities. Designed and trained a convolutional neural network (CNN) to improve robustness under varying imaging conditions and enhance anomaly classification accuracy. The solution reduced human exposure to toxic environments while enabling consistent, scalable, and real-time quality monitoring.

Experience

Aparat Style (Sabaidea Holding) Tehran, Iran

Senior ML Engineer Jan 2025 – Present

AparatStyle is a leading visual discovery and shopping platform for the fashion and home decor markets, using Visual Search (Lens), Virtual Try-On (VTON), and deep textual search to connect users with a catalog of more than 10 million products and enhance the shopping experience.

I was responsible for designing, implementing, and scaling the core AI and data infrastructure powering the AparatStyle platform.

- **MLOps & Data Foundation:** Engineered and deployed a robust shop crawler using Airflow, Celery, and Scrapy to manage and maintain a product catalog of over 10M.
- **Cloud Architecture & CI/CD:** Led the migration and refactoring of microservices into a scalable cloud-native architecture. Managed Kubernetes deployments using GitLab CI/CD and Helm Charts, significantly improving platform reliability and scalability.
- **AI Research & Development:** Drove strategic R&D for next-generation personalized shopping agents.
- **Agentic Chatbots:** Developed LLM-based agentic chatbots, including an Agentic Shop Assistant for conversational discovery and an Agentic AI Stylist for personalized fashion and home decor recommendations.
- **Virtual Try-On:** Implemented a high-performance VTON service using state-of-the-art diffusion models, deploying production-ready microservices capable of handling 30 requests per second.

Technologies: Python, Airflow, Scrapy, Milvus, Claude Code, LangChain, LangGraph

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Iran's National Elites Foundation (Part-time) Tehran, Iran

Senior Machine Learning Engineer Sep 2023 – Aug 2025

At INEF, the team focused on solving complex path planning problems in dynamic, temporal graph environments to improve the efficiency of vehicular navigation systems.

I was primarily responsible for algorithm design, performance optimization, and strategic R&D to shape the next generation of path planning solutions.

- **Performance Optimization:** Engineered and implemented novel algorithms that improved path planning computation speed by 10x, significantly enhancing real-time responsiveness.
- **Dynamic Graph Modeling:** Extended graph models to support temporal and dynamic behavior, incorporating detailed vehicle features and constraints for accurate real-world navigation.
- **Reinforcement Learning R&D:** Led research to identify and evaluate the most effective reinforcement learning methods for adaptive and efficient path planning in complex environments.
- **Simulation & Validation:** Designed and developed a comprehensive simulator to evaluate the accuracy, performance, and real-world viability of path planning algorithms and models.

Technologies: Python, NetworkX, Neo4j, Redis, Docker, FastAPI, Git

Experience

Deed (Tosan Holding) Tehran, Iran

Senior Machine Learning Engineer Jun 2022 - Jan 2025

Deed is an innovative visual platform that enables users to explore millions of inspiring pieces of content, interact with their favorites, create their own, and purchase products seamlessly. The platform uses AI-powered tools to improve discovery, understanding, and engagement.

I was responsible for the end-to-end development, implementation, and deployment of several core AI services for content understanding, moderation, and personalization.

- **Product Identification Model:** Developed a model to identify and localize specific products within images.
- **Image Retrieval System:** Built an image retrieval system to match detected items with the closest corresponding products in online stores.
- **Semantic Search:** Implemented semantic search using both visual features and text descriptions to improve content discovery.
- **Data Pipeline (Crawling):** Built a scalable crawling and monitoring pipeline to maintain data freshness and volume.
- **Personalized Feed Generation:** Designed and implemented a personalized home feed model to improve user engagement and discovery.
- **Weak-Learner Development:** Developed weak learners to capture and categorize fine-grained visual aspects and image features.

Technologies: Python, Pytorch, Transformers, Elasticsearch, MongoDB, PostgreSQL, RabbitMQ, Celery, Apache Kafka, Kubernetes, Git, Haystack

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AIMedic (HiWEB Holding) Tehran, Iran

Lead Natural Language Processing Engineer Jun 2021 – Jun 2022

At AIMedic (HiWEB Holding Company), the team focused on developing chatbot builder software with modules for text preprocessing, model analysis, response generation, and post-processing.

I contributed across the full development lifecycle of the core chatbot platform, helping ensure strong model performance, scalable infrastructure, and specialized language support.

- **Core Language Model:** Developed and integrated a bilingual ALBERT-based language model for handling both formal and informal Persian text.
- **Chatbot Architecture:** Implemented scalable chatbot services on Kubernetes, ensuring high availability and performance.
- **Medical OCR:** Contributed to OCR solutions for processing and digitizing medical corpus data.
- **Information Retrieval:** Developed accurate medical information retrieval systems for complex datasets.
- **Text Mining:** Applied advanced text mining techniques to radiology reports for diagnostic insight extraction and structured data analysis.

Technologies: Python, TensorFlow, Transformers, Git, Ansible, Gitlab CI/CD, Docker, Kubernetes, MongoDB, PostgreSQL, Elasticsearch

Experience

Self-Employed (Contract) Tabriz, Iran
Freelance AI Developer Sep 2016 - Aug 2021
Freelance Machine Learning Engineer | B2B Solutions (5 Years)

Delivered multiple outsourced B2B projects, applying advanced machine learning and computer vision techniques to build enterprise-grade software solutions.

- **Employee Scheduling:** Developed a face recognition system integrated with employee scheduling software for secure and efficient check-in/check-out.
- **Vehicle Monitoring:** Implemented real-time car tracking and parking occupancy detection systems using computer vision.
- **Handwritten Recognition:** Developed robust handwritten text recognition models across various document types.
- **E-Clinic Product Development:** Contributed to e-clinic products by developing detection models for COVID-19 and breast cancer analysis.
- **Deployment & Operations:** Improved delivery speed and reliability by containerizing most products with Docker and establishing automated CI/CD pipelines.
- **Scalable Infrastructure:** Deployed and managed multiple solutions on Kubernetes to ensure scalability and high availability for clients.

Technologies: Python, C++, Bash, TensorFlow, PyTorch, Transformers, Git, Jenkins, GitLab CI/CD, Docker, Kubernetes, PostgreSQL, MongoDB, Redis, ELK, Apache Spark

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OrumSanat Urmia, Iran
AI Developer Sep 2015 – Sep 2016
I was responsible for developing high-performance models for real-time object detection and tracking.

- **Real-Time Computer Vision:** Developed and implemented real-time object detection and tracking models, leveraging OpenCV in a C++ environment for high-performance execution.

Technologies: C++, TensorFlow, OpenCV

Open Source Projects

NaBot Jul 2021 – Dec 2021
NaBot – Medical Chatbot (Rasa-based AI Assistant)
Developed an intelligent medical chatbot on the Rasa framework to provide users with reliable information about medications and laboratory tests through natural language interaction.

- **Conversational AI:** Implemented intent classification and dialogue management to handle user queries effectively.
- **Medical Data Access:** Developed custom actions for retrieving structured medical data, including medication dosage guidance and lab test explanations.
- **Modular Architecture:** Designed the system with separate action and server components to support scalability and easy extension with new medical datasets and features.
- **Deployment:** Enabled simplified setup and production readiness through Docker-based deployment.

Technologies: Rasa, Python, Docker, NLP, REST APIs

Skills

Machine Learning & AI
LLMs, Agentic AI, Retrieval-Augmented Generation (RAG), Recommender Systems, Computer Vision, Natural Language Processing, Semantic Search, Image Retrieval, Reinforcement Learning, Diffusion Models, Virtual Try-On

ML Engineering & MLOps
Production ML Systems, Model Serving, ML Pipelines, Data Pipelines, Web Crawling, Airflow, Celery, Docker, Kubernetes, GitLab CI/CD, Helm, CI/CD, Monitoring, Microservices

Search, Data & Infrastructure
Vector Databases (Milvus & Qdrant), Elasticsearch, PostgreSQL, MongoDB, Redis, Neo4j, Apache Kafka, RabbitMQ

Programming & Tools
Python, C++, Bash, FastAPI, Scrapy, LangChain, LangGraph, Git, Linux

System & Product Strengths
System Design, Scalable AI Architecture, Personalization, Conversational AI, Cloud-Native Deployment, Real-Time Inference, Experimentation & Applied Research

Volunteering

Ubuntu Iranian Team Mar 2017 – Present
Iran
Ubuntu LoCo Team Member & Community Contributor (Iranian Team)
Active contributor to the Ubuntu Local Community (LoCo) Team, supporting the Iranian Ubuntu community in collaboration with global Ubuntu volunteers. Participated in community-driven initiatives to promote, advocate, and improve Ubuntu as a free and open-source operating system.

Contributed to various activities including community engagement on Ubuntu forums, assisting users with technical issues, and encouraging adoption of Ubuntu within the local tech community. Involved in knowledge sharing, translation efforts, and discussions aimed at improving accessibility and usability of Ubuntu for Persian-speaking users.

Supported the broader mission of the LoCo teams by fostering collaboration among enthusiasts, developers, and users to strengthen the open-source ecosystem and promote Linux-based solutions.

Interests

Reading **Persian traditional music** **Playing Chess**

Languages

Persian
Native / Bilingual Proficiency

Azerbaijani
Native / Bilingual Proficiency

English
Advanced proficiency in reading and writing